

SafeExec Final Integrated Submission

AI-use disclosure. Drafted with Codex (GPT-5, Codex desktop app) from the final report, artifact package, evidence bundle, and presentation deck.

AI-drafted, student-revised. Full audit trail: docs/ai-use-log.md.

Assignment: 14 Hard Stop 7: Final Integrated Submission and Presentation/Demo

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Package Summary

This final integrated package submits the SafeExec technical report, computational artifact, evidence bundle, and presentation/demo materials as one coherent capstone submission.

SafeExec is a hardened, threat-modeled Python execution sandbox for LLM-agent tool use. The implementation includes a structured execution model, local development backend, hardened Docker command path, gVisor runsc path, API shell, validation scripts, reproducibility docs, and final artifact package.

Final Submission Contents

Required item	Submitted file(s)
Final report	Final-Technical-Report.docx, Final-Technical-Report.pdf, final-technical-report.md
Computational artifact package	safeexec-final-artifact-package.zip
Final evidence bundle	final-evidence.zip, evidence/
Presentation/demo deck	Final-Presentation-and-Demo-Deck.pptx, Final-Presentation-and-Demo-Deck.pdf
Walkthrough/demo script	final-presentation-demo-script.md
Release/archive notes	final-release-notes.md
AI-use disclosure	docs/ai-use-log.md

Technical Result

Fresh final evidence passed:

- smoke execution,
- unit/functional tests,
- local validation/API trace,
- reproducibility audit.

The strongest Docker/gVisor target-host evidence remains:

Backend	Passed / Total	Pass rate
Local subprocess	30 / 30	100.0%
Hardened Docker	50 / 50	100.0%
gVisor (runsc)	49 / 50	98.0%

Main Limitation

The final package is honest about the largest remaining gap: the HumanEval/MBPP functional subset and expanded adversarial suite are not yet at the original stretch-target scale. The submitted artifact is still runnable, reproducible, and evidence-backed, but it should not be described as a production sandbox or as proof of general adversarial safety.

Reviewer Navigation

1. Read `Final-Technical-Report.pdf`.
2. Inspect `final-test-evidence-appendix.md` for the evidence map.
3. Open `safeexec-final-artifact-package.zip` for source, tests, scripts, reproducibility docs, and evidence files.
4. Review `Final-Presentation-and-Demo-Deck.pdf` or `.pptx`.
5. Use `final-presentation-demo-script.md` to record or review the walkthrough.